COMPUTER VISION	
ASSIGNMENT-5	
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Manual Count vs Automated Count Evaluation

Automated Counting Results

The implemented system uses YOLO for person detection, ByteTrack for object tracking, and Supervision for detecting line-crossing events.

The automated system produced the following results:

Measurement	Automated Count	Manual Count
Entry (IN)	16	18
Exit (OUT)	15	10
Total Crossings	31	28

Analysis of Results

The automated system detected 16 entries and 15 exits, while the manual observation recorded 18 entries and 10 exits.

The automated system undercounted entries by 2 and overcounted exits by 5. Overall, it recorded 31 crossings compared to 28 manually observed crossings, resulting in a difference of 3 crossings.

Possible Reasons for Differences

The differences between manual and automated counts may be caused by:

1. **ID Switches:** The tracking algorithm may assign a new ID to an existing person, potentially affecting crossing counts.
2. **Group Crossings:** When multiple people cross the virtual line together, the system may miss or incorrectly register individual crossings.
3. **Line Placement:** The position of the virtual line may affect how accurately crossing events are detected.
4. **Detection Failures:** YOLO may temporarily fail to detect people due to occlusion or low visibility.

Conclusion

The automated system successfully implemented pedestrian detection, object tracking, and virtual line-crossing counting. However, the comparison shows that its results differ from manual observations.

These differences highlight the challenges of accurate pedestrian counting and demonstrate the importance of proper tracking, line placement, and detection reliability in computer vision systems.